

Classroom Problem-Solving Activity*Building a Simple Java GUI: Movie Night Picker*

Course topic	Graphical User Interfaces (java.awt / javax.swing)
Concepts practiced	JFrame, JPanel, JRadioButton + ButtonGroup, JCheckBox, JList, JButton, ActionListener, ListSelectionListener
Format	Group work, in-class design + coding activity
Duration	45–50 minutes
Discussion	15-20 minutes

1. Problem Statement

A small cinema wants a one-screen kiosk where a customer picks a movie, a showtime, and whether they want popcorn, then presses one button to see their order confirmed.

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Movie Night Picker
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Movie:      Showtime:
[ Action Hero ] [ 5:00 PM ]
[ Comedy Night ] [ 8:00 PM ]
[ Space Quest ] [ ] Add Popcorn
               [ Confirm Order ]
You picked: _____
  
```

2. What You Need to Build

1. A JFrame titled "Movie Night Picker" with a JPanel added to it.
2. A JList of three movie names, single selection only.
3. Two JRadioButton objects (5:00 PM / 8:00 PM) in one ButtonGroup so only one can be chosen.
4. One JCheckBox labelled "Add Popcorn".
5. One JButton labelled "Confirm Order".
6. One JTextField (not editable by the user) to show the result.
7. An ActionListener on the button that reads the list selection, the chosen radio button, and whether the check box is ticked, then writes one summary sentence into the text field.
8. If no movie is selected when the button is clicked, show a message such as "Please choose a movie" instead of letting the program crash.

3. Discussion Questions

1. Why do the two showtime buttons need to be in the same ButtonGroup?
2. Why can you use one ActionListener to read the JList, the radio buttons, and the check box all at once, even though they are different component types?
3. What does `getSelectedValue()` return if nothing in the list has been clicked yet, and why must your code check for this?